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Comparative study and analysis on skin cancer detection using machine learning and deep learning algorithms

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Abstract
Exposure to UV rays due to global warming can lead to sunburn and skin damage, ultimately resulting in skin cancer. Early prediction of this type of cancer is crucial. A detailed review in this paper explores various algorithms, including machine learning (ML) techniques as well as deep learning (DL) techniques. While deep learning strategies, particularly CNNs, are commonly employed for skin cancer identification and classification, there is also some usage of machine learning and hybrid approaches. These techniques have proven to be effective classifiers of skin lesions, offering promising results for early detection. The paper analyzes various researchers' reviews on skin cancer diagnosis to identify a suitable methodology for improving diagnostic accuracy. A publicly available dataset of dermoscopic images retrieved from the ISIC archive has been trained and evaluated. Performance analysis is done, considering metrics such as test and validation accuracy. The results indicate that the RF(random forest) algorithm outperforms other machine learning algorithms in both scenarios, with accuracies of 58.57% without augmentation and 87.32% with augmentation. MobileNetv2, ensemble of Dense Net and Inceptionv3 exhibit superior performance. During training without augmentation, MobileNetv2 achieves an accuracy of 88.81%, while the ensemble model achieves an accuracy of 88.80%. With augmentation techniques applied, the accuracies improved to 97.58% and 97.50%, respectively. Furthermore, experiment with a customized convolutional neural network (CNN) model was also conducted, varying the number of layers and applying various hyperparameter tuning methodologies. Suitable architectures, including a CNN with 7 layers and batch normalization, a CNN with 5 layers, and a CNN with 3 layers were identified. These models achieved accuracies of 77.92%, 97.72%, and 98.02% on the raw data and augmentation datasets, respectively. The experimental results suggest that these techniques hold promise for integration into clinical settings, and further research and validation are necessary. The results highlight the effectiveness of transfer learning models, in achieving high accuracy rates. The findings support the future adoption of these techniques in clinical practice, pending further research and validation.

Keywords Melanoma · Benign · Machine learning · Deep learning · SVM · KNN · Random forest · CNN · VGG16

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1 Introduction

Healthcare knowledge extraction is a highly difficult task due to issues, like noisy and unbalanced datasets which come from clinical investigations, and are generally uncertain and variable. Recently, a variety of machine learning algorithms have been reviewed and implemented to assess their potential for usage in the medical industry. Typically, the efficiency of the techniques is compared to the actual diagnostics of the medical domain experts professionals that specializes in diagnosing particular diseases to provide an efficient methodological evaluation of classifiers using performance indicators. Accuracy, sensitivity, and specificity are the three performance metrics that make up the confusion matrix for any algorithm that is implemented [82].

Diseases, like COVID, pneumonia, heart problems, brain tumors, and skin cancer, can be detected using ML and DL algorithms. Certain diseases require image input for detection. In [6, 9], the lightweight hybrid deep learning approach (CNN + LSTM) is implemented for cardiac vascular disease classification. The different types of spectrum classification using ECG heart beats have been analyzed using CNN architectures [9]. Chest X-ray images were trained by hybrid approach (CNN + SVM, KNN, RF, Softmax) to classify the normal based and pneumonia features [71]. Using PCG signal, AOCNet was capable for classifying the heart sounds normal or abnormal [8]. In [7] CAD, categorization was implemented using SVM in digital for detection of the breast cancer.

Unchecked melanocyte development is the cause of melanoma [83]. Factors such as symmetry, color, size, and shape play a crucial role in detecting skin cancer and distinguishing between benign and malignant lesions [83]. Many countries worldwide, especially the United States, have reported increasing death rates from skin cancer [70]. Recent cancer statistics reveal an estimated 1.9 million new cases of this type of cancer, with an anticipated 608,570 deaths in the United States alone [17]. Early diagnosis is crucial in reducing the mortality rate. Exposure to sunlight is strongly associated for developing both malignant and benign type of skin cancer [70]. Current classifications include malignant melanoma and benign, including BCC and SCC as shown in Fig. 1 [70].

Types of Malignant Skin cancer:

Melanoma (Fig. 1a)**:** Melanoma, the most severe malignant tumor, begins in the melanocyte cells of the epidermis. The tumor spreads quickly, has a high fatality rate due to early metastasis, and is challenging for treatment. Although it only accounts for 4% of all skin cancers, it results in death in 80% of cases. However, if detected early, it has a 95% cure rate, highlighting the significant impact of early diagnosis on improving survival prospects [69].



(a)MEL (b) SCC (c) BCC

Fig. 1 Skin cancer types [70]

Squamous cell carcinoma (Fig. 1b: The outermost layer of the epidermis, known as squamous cells. Early detection makes it easily treatable, but if left untreated, it can penetrate the skindeeper layers.
Basal cell carcinoma (Fig. 1c): The basal cells, which reside in the innermost layer of the epidermis, are where around 80% of cases begin. Since basal cell development is slow, BCC is often highly treatable and causes minimal harm if detected and treated promptly.

Melanoma, considered a severe type of skin cancer due to its rising incidence rate. Timely detection is of utmost importance in maximizing curative outcomes and ensuring optimal recovery, as failure to identify it promptly can lead to fatal consequences [65]. Determination of skinlesion classes is important for assessing the stage of the disease and evaluating the probability of being at risk.

The following are the skin lesion classes.

1. BACC -Basal cell carcinoma
2. SEK- Seborrheic keratosis
3. ACK- Actinic keratosis
4. DEF-Dermato fibroma
5. MEL -Melanoma
6. NEVI -Nevus
7. VASCL-Vascular lesion

Dermatologists have historically utilized various approaches, such as the ABCD rule, three checklists, seven checklists, and the Menzies rule, to detect and diagnose skin cancer. However, these dermoscopic imaging procedures are susceptible to errors and require extensive expertise. Detecting skin lesions using traditional methodologies involves employing common techniques like thresholding, clustering, edge-based analysis, and regional approaches [29, 65] (Table 1).

The aforementioned approaches were found to lack sufficient prediction accuracy. As a result, the researchers are in need of an alternative classification and detection scheme for cancer cells. The system flow is illustrated in Fig. 2. The comparative analysis of skin lesion using ABCD rule has been shown in Table. 2.

Pre-processing techniques are applied to mitigate picture artifacts and enhance lesion visibility. The hair removal approach comprises two main steps: hair restoration and hair detection, employing morphological operations, thresholding, and filtering methods like Gaussian, median, and middle filters [66].

Color calibration is a crucial step in image enhancement to restore accurate lesion colors. Additional image enhancement methods involve adjusting lighting conditions, enhancing contrast, and improving edge sharpness. Illumination reflectance techniques, Retinex algorithms [100], and filters such as bilateral filtering are commonly utilized to handle illumination variations. Contrast augmentation typically involves a combination of histogram equalization, adaptive histogram equalization, and sharp masking. Edge detection is performed using PCA (principle component analysis). Feature extraction can be achieved manually or automatically using machine learning methods. The ABCD rule methodology relies on feature extraction to facilitate effective skin lesion classification by dermatologists [100].

When applying ML techniques, without any prior knowledge of the issue, learnt features are automatically constructed from datasets. There are several options available, ranging

Table 1 Criteria used by dermatologists for diagnosing melanoma

Approaches	Description	Methodology
ABCDE [29]	The rule is based on morphological characteristics that encompass asymmetry, irregular edges, non-homogeneous coloration, and a diameter equal to or greater than 6 mm. It also considers the concept of evolution, which involves assessing changes over time in parameters such as shape, size, color, elevation, and the presence of symptoms like itching or bleeding.	1. Acquisition of Image 2. Preprocessing (PP) 3. Image segmentation 4. Selecting Feature using ABCDE rule 5. Classification
7 Point Checklist [43]	Each of the minimal criteria, namely pigmentation, irregular streaks, and irregular patches, can be assigned a score. Additionally, the major criterion of abnormal pigment networks can also be assigned a score.	1. Image processing techniques 2. Pattern recognition
Menzies Method [73]	There are a total of 11 features that are evaluated as either present or absent, including two negative features and nine positive features.	Positive characteristics of melanoma include: Multiple brown dots scattered throughout a blue-white veil, focal pseudopods, scar-like depigmentation, and five to six different coloured black spots or globules on the periphery.

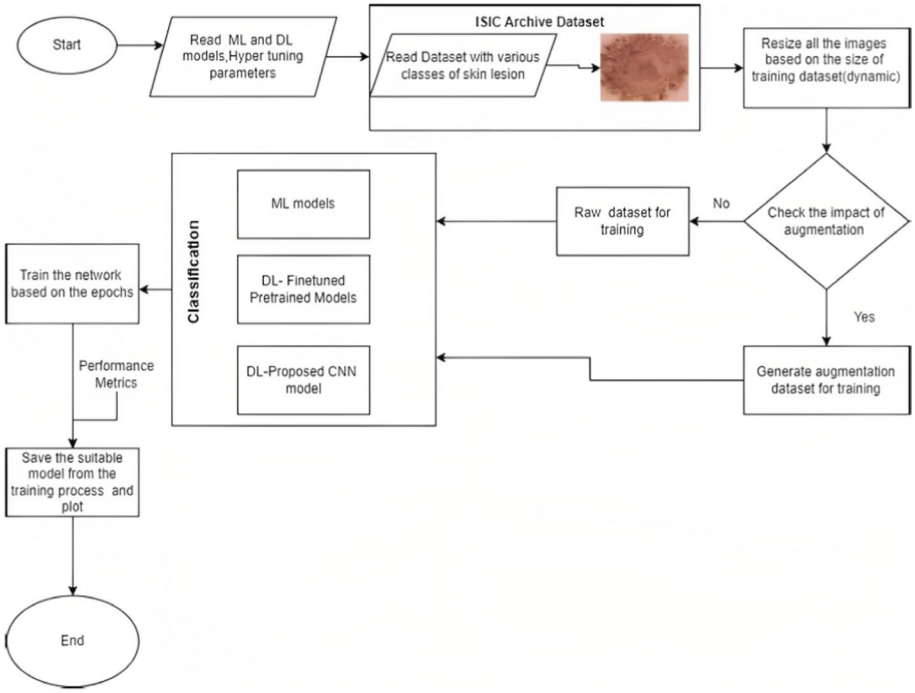


Fig. 2 System architecture

from traditional methods to DCNN. In this review article, multiple methods are investigated to determine their performance in classifying skin lesions and detecting skin cancer. Numerous studies have demonstrated that deep learning techniques exhibit superior efficiency compared to skilled dermatologists when it comes to identifying and detecting skin lesions. However, it is important to evaluate these methods that fall outside their training scope. Automatic dermoscopic image classification encounters various difficulties, including insufficient distinguish between the lesion and normal skin, similarities in appearance between benign and malignant cases, and difference in individuals' conditions of skin. In relation to the latter point, skin tone is an important but often overlooked factor [54].

The available datasets for training machine learning algorithms primarily consist of dermoscopic images of individuals with pale skin. However, there is a need to address this gap and expand the datasets to ensure accurate performance in identifying skin lesions in individuals with dark skin.

The utilization of algorithms offers several advantages. Firstly, it provides an opportunity to avoid unnecessary biopsies and ensure timely diagnosis of melanomas. Additionally, these techniques enable the diagnosis of skin conditions without physical contact. This not only enhances patient comfort but also reduces the cost associated with non-melanoma skin cancer diagnosis and treatment, which has been shown to be expensive.

There are multiple sections in this article. The review's methodology was covered in Section 2 of the article. In Section 3, several techniques (ML and DL), and hybrid AI strategies are analysed. Review of ML/DL algorithms is the primary topic of Section 4. In Section 5, an experimental analysis on deep learning models has been conducted.

Table 2 Comparative analysis of skin lesion using ABCD rule

Ref	Diagnosis	Dataset	Technique	Proposed methodology	Accuracy	Drawbacks
[66]	2 class (Benign and Melanoma)	200 dermoscopic images	ABCD rule	1. Preprocessing: Gabor filters were used to pre-process the image by detecting the hair 2. Geodesic Active Contours: To detect the lesion boundaries	Sensitivity- 91.25% Specificity-95.83% Overall Accuracy-94%	Possible for misclassification Bad contour detection Inability to detect structure Can't work with large dataset.
[100]	2 class (Benign and Melanoma)	ISIC 2016, 2017, 2018 and PH2	ABCD	1. Preprocessing: Unwanted hair in the skin lesion were removed using median filters. 2. LSM (Lesion segmentation module): To divide the area of interest (ROI) 3. BDM (Border Detection Module): A segmented image input into the BDM helps to identify the lesion area's single pixel border. 4. Utilize CIEM to extract the color regions from the lesion (Color Information Extraction Module).	Sensitivity-97.69% Specificity-97.97% Overall Accuracy-97.86%	Trained and Tested separately rather than combining the dataset. In this article the proposed system works on separate dataset.

Section 6 addresses the research challenges. Section 7 consists of discussions, conclusions and future enhancements based on the review.

2 Methodology

2.1 Searching strategy

An extensive literature review was conducted, focusing on predefined endpoints to determine the appropriate technique for skin cancer detection. To select relevant articles, specific keywords such as Melanoma, dermoscopic images, machine learning, deep learning, and melanoma classification were used in combination with "and" and "or" operators in electronic database searches. The inclusion of studies and extraction of relevant data followed the preferred reporting guidelines for systematic reviews. All relevant information was obtained from reputable sources and meticulously analyzed for further analysis. The completion of this survey will provide insightful, logical, and robust solutions to the research questions at hand.

2.1.1 Structure of study

To delineate the study, the initial phase involves assessing the system using a three-layered approach.

- The first layer is the planning stage, where the overall framework and objectives of the study are established. This includes defining the research questions, identifying the scope of the study, and outlining the methodology to be employed.
- The second layer focuses on data selection and evaluation. In this stage, relevant datasets are identified and collected for analysis. The quality and suitability of the data are carefully assessed, considering factors such as sample size, data accuracy, and representativeness.
- The third layer encompasses result generation and conclusion. Here, the gathered data is analysed through the use of suitable statistical and computational techniques to produce insightful findings and conclusions. The results obtained are rigorously evaluated and interpreted, taking into account the research objectives and hypothesis. Finally, a comprehensive conclusion is drawn based on the findings, highlighting the implications and potential avenues for future research.

This three-layered approach provides a systematic and robust analysis.

2.1.2 Research problems

Framing the study questions is crucial to conduct a focused and relevant review on the specific topic. The following inquiries have been formulated based on the most recent systematic review:

- What methods are employed in for skin cancer identification?
- What methods are utilized in DL for the skin cancer diagnosis?
- What are the key data points required for accurate identification of skin cancer?
- How can the most appropriate learning model for prediction be determined?

By addressing these research questions, the study aims to explore and analyze the various methodologies used in both ML and DL approaches for skin cancer identification. Additionally, the investigation aims to identify the essential data points required to enhance the accuracy. Lastly, the study seeks to find the learning method suits to do effective prediction in the context of diagnosis. Through a comprehensive survey of the existing literature, valuable insights and recommendations can be provided in response to these important research questions.

2.1.3 Selection and evaluation criteria

A comprehensive search was conducted to gather relevant research articles for this review. A total of 150 full-text articles were examined from reputable databases including Web of Science, Scopus, and PubMed. After the abstract screening process, 115 studies were selected for further analysis. The selection criteria focused on recent advancements in melanoma identification using learning techniques, the visibility and impact of the contributions, as well as the publication venue (high-ranking conferences and journals) and citation count. The chosen studies represent the most significant and representative research on learning-based melanoma detection, primarily published between 2015 and 2021, with a focus on the years 2018 to 2021. The quality of the selected research papers has been carefully evaluated to ensure their relevance and reliability. This evaluation includes considering factors such as journal reputation and citation metrics. The process adhered to the PRISMA guidelines for systematic review and meta-analysis, as depicted in Fig. 3 [59]. This diagram shows the study selection process and inclusion/exclusion criteria, ensuring transparency and rigor in the review methodology.

2.2 Skin lesion –public databases

Computerized systems play a crucial role in diagnostics, especially in melanoma. In this section, the dataset used for skin cancer detection approaches has been discussed [47]. The datasets serve as a valuable resource for training and evaluating various algorithms and techniques. The diagnostic models performance can be analyzed by dermoscopic images that enable researchers to assess accurately.

2.2.1 ISIC archive

ISIC collection offers a wide range of datasets for cutaneous lesions. One of the prominent datasets is the ISIC2016. It consists of 900 and 379 dermoscopic images of training and testing subset. ISIC2017 collection, the dataset is divided into three categories: benign nevi, benign SK, and melanoma, with a total of 2000 images. Among these, 150 images for validation, and 600 are designated for test process [47].

Moving on to ISIC2018, the dataset includes seven classes of skin lesions along with metadata, training and testing input images [22]. The categories in the ISIC2019 dataset encompass 8 classes BACC, SEK, ACK, DEF, MEL, NEVI, VASCL, and SQCC. The test data for ISIC2019 consists of 8239 images with outlier classes. [22]. To enable comprehensive skin cancer detection, modern methods should be capable of accurately identifying images from these diverse categories. Additionally, it has the metadata information for analysis and research purposes.

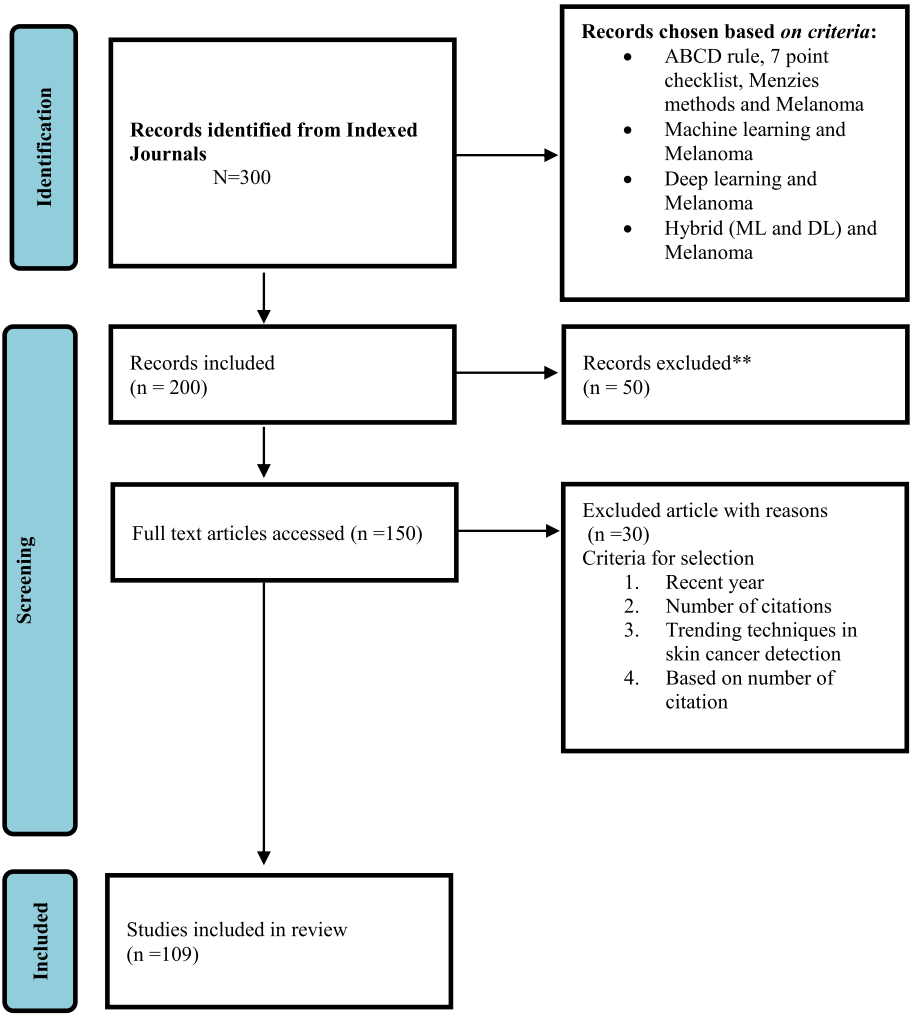


Fig. 3 PRISMA Methodology-systematic review on algorithms and melanoma identification [59]

2.2.2 HAM10000

The human-versus-machine dataset for skin lesions, known as HAM10000, consists of 10,000 training photos [95]. This freely accessible dataset is an important resource for study and advancement in the field of skin lesion classification. The dataset were contributed by the Cliff Rosendahl's research and the skin cancer center in Queensland, Australia [94].

The images were then edited manually and stored with dimensions of 800 to 600 pixels. These standardized images ensure consistency and compatibility for analysis and evaluation purposes within the dataset.

2.2.3 PH^2

The rise in melanoma cases has spurred the computer-aided diagnosis techniques development in the dermoscopic images classification. The PH2 dataset was created and serves as a benchmarking tool for research purposes. The dataset was collected at the Pedro Hispano Hospital Dermatology Centre in Portugal. The PH2 dataset consists of 200, with 80 images representing common nevi, and another 80 images depicting rare nevi, and 40 images showcasing melanoma skin cancer [72].

Each image in the dataset is accompanied by medical annotations, including lesion segmentation, clinical diagnosis, and assessment of various dermoscopic criteria such as colors and streaks [1]. These annotations provide valuable information for training and algorithms evaluation for the contribution of skin lesion classification techniques.

2.2.4 Med node

The Med node systems macroscopic images were developed and tested using a dataset consisting of 100 nevus photos and 70 melanoma images. The dataset was obtained from the department of dermatology at the University Medical Centre Groningen (UMCG) [36]. These images were utilized to evaluate and validate the performance of the Med node systems macroscopic image analysis techniques. By using this dataset, the researchers were able to assess the effectiveness and accuracy of their proposed methods in differentiating between nevi and melanoma, contributing to the advancement of skin cancer diagnosis and classification.

Table 3 contains NEVI/AN (Atypical NEVI), COM NEVI (Common Nevi), MEL (Melanoma), SEK (Seborrheic keratosis), BACC (Basal cell carcinoma), DEF (Dermato fibroma), ACK (Actinic keratosis), VASCL (Vascular lesion), SQCC (Squamous cell carcinoma).

3 AI techniques

Artificial intelligence (AI) is the machine that carries out tasks need the intelligence of people, such as learning, problem-solving, planning, and interpreting natural language. AI can be divided into two main branches, each with its own characteristics. The first branch is transfer learning (TL), which involves leveraging knowledge from pre-trained networks to improve performance on new tasks or domains. TL allows the transfer of learned features or representations from one task to another, reducing the need for extensive training on new datasets. It enables the adaptation and reusability of models, making AI systems more efficient and effective in handling various tasks [38].

3.1 ML algorithms

The four crucial steps in the ML method are pre-processing, feature extraction, segmentation, and classification to categorise skin lesions. Pre-processing involves cleaning up the input data by removing noise and artefacts. Segmentation involves identifying and delineating the lesion region in the image. Feature extraction is a crucial step where relevant information is extracted from the segmented region. In the context of machine learning, domain-specific features are commonly used. These features capture

Table 3 Publicly available skin lesion dataset summary

Dataset	Skin lesion classes									Total. no. of images
	NEVI/AN	COM NEVI	MEL	SEK	BACC	DEF	ACK	VASCL	SQCC	
ISIC2016 [48]	726	–	173	–	–	–	–	–	–	899
ISIC2017 [49]	1372	–	374	254	–	–	–	–	–	2000
ISIC2019 [50]	12,875	–	4522	2624	3323	239	867	253	628	25,331
ISIC2020 [51]	27,124	5193	584	135	–	–	–	–	–	33,126
HAM10000	6705	–	1113	1099	514	115	327	142	–	10,015
PH^2	80	80	40	–	–	–	–	–	–	200
MED NODE	100	–	70	–	–	–	–	–	–	170

important characteristics of the lesion that can be utilized in machine learning algorithms. The next stage involves training the machine learning model using a dataset that includes both image features and corresponding labels. Several methods, including LR (linear regression), LOR (logistic regression), KNN, SVM, and RF are frequently used in dermatology for prediction. SVM is particularly useful for classifying data into different groups by searching for hyperplanes. RF generates an ensemble of randomly constructed decision trees to determine the most frequent outputs. KNN is utilized for data classification and retrieval based on the nearest neighbors. Each of these strategies will be detailed in subsection.

3.1.1 SVM

SVM is a powerful ML model, capable of performing regressions, outlier identification, and linear and nonlinear classifications using kernel techniques. SVMs are particularly effective when dealing with complex datasets, although they may not perform optimally on extremely large datasets. One of the key features of SVMs is their ability to create hyperplanes to solve classification problems [23]. These hyperplanes separate different classes or categories in the data space, allowing for accurate classification and prediction. SVMs have been widely used in various domains due to their flexibility and ability to handle both linear and nonlinear relationships in the data.

3.1.2 DT (decision trees)

DT (Decision trees) are flexible methods that do not rely on assumptions about data distribution or classifier design. They can be applied to both categorical and numerical variables, making them suitable for various tasks such as classification, regression, and handling multiple outputs. Decision trees provide precise and effective classifications, even for large and complex datasets. RF (Random forest) builds upon decision trees by creating an ensemble model. RF combines multiple decision trees, each with high variance, to form a more robust model with improved generalization performance. By leveraging the diversity of individual trees, random forests enhance the accuracy and stability of predictions [24].

3.1.3 KNN

The KNN technique categorise raw data according to how closely it resembles nearby labelled data. The classifier's parameters, such as K value, indicates the number of closest neighbors to take into account, are established. The new data's label is chosen by a significant majority vote among its K closest neighbours. Distance measures like the manhattan distance are used to calculate the data points separation. To prevent overfitting or underfitting, a value of K between 3 and 10 is commonly selected. This range strikes a balance between including enough neighbors for accurate classification and mitigating the impact of outliers or noise [11].

3.1.4 ANN

The approach developed from discussions surrounding the brain aimed to tackle challenges as intricate as those encountered in the real world. Initially, attempts were made to simulate the human brain and its synaptic connections through neural networks. However, due to a limited understanding of the brain's complex operations, these attempts were not successful. Instead, Artificial Neural Networks (ANNs) were constructed using simpler, organized functional units known as neurons or nodes. These neurons are interconnected through simulated connections and organized into layers. The layers, which make up the neural network's central structure, extract and process information that is pertinent to the current circumstance.

3.2 DL (deep learning)

A sophisticated algorithm called DL (Deep learning) uses a complicated architecture that closely matches the structure of the human brain to be able to independently learn from a dataset. This technique finds applications in medical image analysis, where it aids in prediction, diagnosis, and detection tasks. Compared to traditional methods, current neural networks used for image recognition exhibit superior performance in skin lesion classification [2]. The initial step involves in receiving the image and understanding its class characteristics. However, the way humans recognize images significantly differs from that of machines, which perceive images as pixels. In deep learning, two prominent techniques, namely ANN (Artificial Neural Networks) and CNN (Convolutional Neural Networks), are widely utilized. These neural networks are specifically designed for image detection and computer vision challenges [28]. Unlike traditional machine learning approaches, CNN utilizes a sequence of convolution, pooling, and non-linear layers to process and pre-process the image as a two-dimensional vector, ultimately producing output through the Fully Connected Layer (FCN) (http://www.datamind.cz/cz/Sluzby-Data-Science/umela-intelingence-AI-ML-machine-learning-neural-net). The overall process of classification using DL is illustrated in Fig. 4. The underlying methodology of deep learning surpasses that of traditional machine learning algorithms in terms of effectiveness and performance.

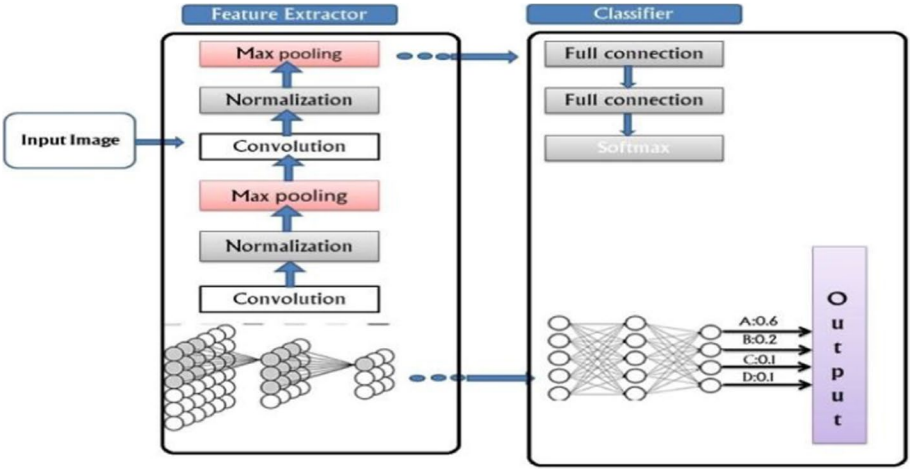


Fig. 4 DL (deep learning) – classification process

3.2.1 CNN

In the detection of melanoma, the choice of classification technique plays a crucial role, and CNN (Convolutional Neural Network) has proven to be the appropriate architecture for image recognition [53]. CNN consists of hidden layers that include convolutional, non-linear pooling, and FC layers [53]. The convolutional layers perform multiple convolutions and are connected to the FC layers [53]. However, the layer that deeply connected is not suitable for melanoma detection as it can lead to overfitting, especially when the lesion images lack distinct features [52]. To overcome the overfitting problem, CNN is well-suited for identifying and classifying images based on different classes [52]. The architecture of CNN can be visualized in Fig. 5.

Convolution layers Convolutional layers in CNN have the ability to capture local patterns, which leads to two significant advantages: they are translation-invariant, meaning the learned patterns can be detected regardless of their position in the image, and they capture spatial hierarchies of patterns. As the depth of the network increases, CNN becomes proficient at detecting more complex visual concepts. In the convolutional process, these layers apply a set of filters to the input image, performing convolutions and producing feature maps that are then passed to subsequent layers for further processing. This hierarchical feature extraction allows CNN to effectively analyze and understand visual information.

Pooling layers CNN uses subsampling, also known as pooling, as a method to make feature maps less dimensional while preserving crucial data. It aids in reducing the network's computational burden and parameter count. Although these filters do not have trainable parameters, they still perform convolution on the input feature maps. To downsample the active maps by aggregating the values in each pooling window (e.g., taking the maximum or average value) and selecting the most relevant features are possible in pooling. Downsampling decreases the spatial pixel values of the active maps while retaining important attributes for subsequent layers.

Batch normalization layer These layers, known as normalization layers, apply a specific function to normalize the input data. They are typically used in the forward propagation stage of the network and do not have any trainable parameters. Their purpose is to ensure that the input data is standardized and falls within a certain range, which can help improve the training process. The usage of these normalization layers has been decreased nowadays, and

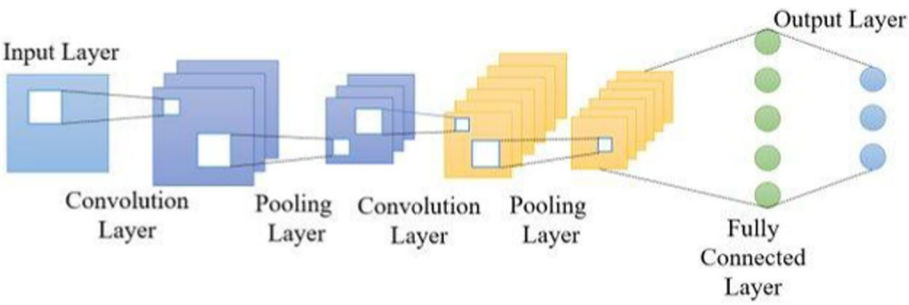


Fig. 5 CNN – basic architecture

alternative techniques such as batch normalization have gained more popularity. Batch normalization allows for normalization within mini-batches during training and has been shown to be more effective in improving the training speed and stability of deep neural networks.

Regularization These layers, known as dropout layers, are used to combat overfitting in neural networks. During each training session, a certain percentage of neurons in these layers are randomly "dropped out" or ignored. This indicates that they don't contribute to the network's forward or backward propagation because their outputs are momentarily set to zero. By randomly dropping out neurons, the network becomes less reliant on specific neurons or connections, and it compels the surviving neurons to acquire more reliable and comprehensive data representations. A common regularisation method that helps avoid overfitting is dropout by reducing the network's tendency.

Fully connected layers Dense layer neurons, often referred to as the FC layer, are combined with every neuron in the previous layer. This means that the activations from the previous layer serve as inputs to every neuron in the FC layer. The FC layer acts as a classifier in a CNN architecture, taking the output feature maps from the previous convolutional or pooling layers and transforming them into a form that is suitable for making predictions or classifications. The last FC layer in the CNN is often referred to as the classifier because it produces the final output of the network, to determine the predicted class.

3.2.2 Pre-trained models

Pre-trained models are existing models developed by others to address similar problems. They have already been trained on large datasets and learned to recognize visual patterns and features. Using pre-trained models, training time and computational resources are reduced. Instead of building a model from initial level, we can adapt the trained model by fine-tuning or retraining only a few layers to suit our specific problem. This approach benefits to speeds up the training process.

3.2.3 Transfer learning

TL is a powerful method that utilizes pretrained CNN models, such as those trained on ImageNet, to achieve good results in a target domain. By reusing the learned weights and representations, the model can adapt to a specific task with limited data. The approach involves freezing the initial layers to preserve generic features and selectively fine-tuning specific layers to capture domain-specific characteristics. This enables efficient adaptation and improves performance on the target task. To fine-tune a pre-trained model, you can follow these steps:

1. Load the pre-trained model and its weights.
2. Replace the last layers with new ones suitable for your task.
3. Freeze the initial layers to retain learned representations.
4. Train the modified model using your dataset, updating only the new layers.
5. Optionally, unfreeze and fine-tune some initial layers if your dataset is large and different.
6. Continue training to refine the model's weights and capture dataset characteristics.

This approach allows you to leverage pre-trained weights and adapt the model to your specific task, even with limited data. Experiment with different hyperparameters for optimal results.

3.2.4 Traditional learning vs deep learning

In traditional method, the initial steps involve data collection and preprocessing, where the dataset is processed and relevant features are extracted [64]. Feature engineering is a crucial part of this process, where domain expertise is applied to manually train the dataset and identify important predictive features. This approach requires human programming and can be time-consuming [103]. DL models, leverage predefined architectures and do not rely heavily on feature engineering. They can automatically learn and make predictions without extensive manual intervention. This makes deep learning more accurate and efficient compared to traditional machine learning methods [20]. Figure 6 provides a visual representation of TL and DL.

4 Review on ml and dl algorithms

In order to improve melanoma categorization, segmentation, and analysis, the literature review on skin lesion classification highlights the major application of ML and DL approaches in dermoscopic imaging. DNN are trained on large datasets of benign and malignant melanoma images with expert annotations. Several datasets, including ISIC archive (2018, 2019, 2020) are commonly used in the research [99].

The reviewed articles cover various approaches, starting with those that primarily focus on machine learning (ML) methods for skin cancer diagnosis. Subsequently, there are publications that specifically emphasize deep learning (DL) methods. Finally, there are publications that integrate methods with ML and DL to evaluate the models' performance using evaluation metrics. The performance indicators provide insights into accuracy and reliability of the classification models [AUC].

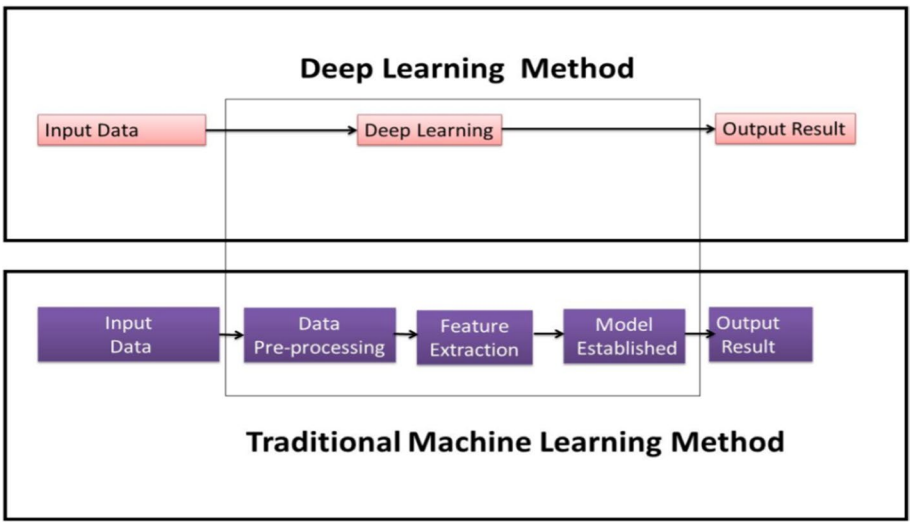


Fig. 6 Traditional machine learning versus deep learning

4.1 Machine learning strategies

In Yuan et al. [105], the author focused on classifying the benignity and malignancy of skin lesions using texture data. The study employed a three-layer support vector machine (SVM) mechanism, achieving an accuracy of 86.9%. The algorithm's performance was validated using 22 pairs of genuine clinical images.

The authors in [35] explored the use of color images for melanoma detection. The proposed approach involved preprocessing steps such as background light correction and iterative dilation-based noise removal. The generated image was then used as input for SVM classification. Sequential minimal optimization was used to optimize SVM parameters. The results indicated promising outcomes in accurately categorizing melanoma samples.

Babu et al. [13] focused efficient diagnosis of skin cancer using HOG features and SVM. The ISIC 2018 dataset was preprocessed by converting it into a grayscale image using median filters. HOG features were extracted from the preprocessed image and used for SVM-based classification. The framework achieved a precision of 76%.

Adjed et al. [3] employed the LBP method for classification. LBP was used to extract local texture information to create statistical characteristics that can differentiate between melanoma and non-melanoma skin tissues. A database was prepared and trained based on color and texture variables derived from the images. The SVM classification was performed based on these variables, achieving an accuracy of 76.1%.

Fei et al. [33] utilized SVM on PSO (particle swarm optimization) to maintain accuracy. The study utilized the PH2 dataset, which consisted of 200 images and 46 texture features for full image analysis and classification.

These studies demonstrate various ML techniques, such as SVM, HOG features, LBP, and PSO, for skin cancer classification, highlighting their potential in improving diagnostic accuracy.

The comparative survey on detecting skin cancer using ML algorithms is shown in Table 4.

Preprocessing (PP), Feature Extraction (FE), Feature Selection (FS),Classification (CLF), Segmentation (SEG), Support vector machine (SVM), K-nearest neighbor (KNN), Random forest (RF), Decision Tree (DT), Naïve Bayes (NB),Binary Harris Hawks optimization algorithm (BHHO).

4.2 Deep learning strategies

CNN models have shown exceptional effectiveness in diagnosis in the realm of medical imaging, classification, and segmentation tasks [39]. DeVries and Ramachandram [25] developed a training framework using the ImageNet dataset, where a multi-scale CNN architecture was employed. Their approach involved fine-tuning a pretrained v3 deep neural network using lesion images.

A technique for features extraction from skin lesions was put out by Mahbod et al. [67] using pretrained models like AlexNet, ResNet-18, and VGG-16. An SVM classifier was then trained using the retrieved characteristics. When tested against the 2017 ISIC dataset, their model has a 97.55% accuracy rate.

Dorj et al. [27] utilized pretrained deep CNNs, particularly AlexNet, for categorizing various types of skin lesion images.

Table 4 Machine learning approaches on skin cancer detection

Ref	Diagnosis	Dataset	Classifier	Methodology	Accuracy
[46]	Two class Benign /Malignant	HAM10000-2000 Samples	SVM	1. PP-K-Means Clustering 2. SEG- LBP, HOG, BOVW 3. CLF-SVM	99.8%
[69]	Two class Cancer/Non Cancer	PH^2	SVM (Linear and Radial kernel function)	1. PP-Gaussian Filter 2. SEG- Otsu 's thresholding 3. CLF-SVM (Linear and Radial Kernel function)	92.30%
[31]	Risk level -low, medium and high	DermIS, DermQuest	SVM, Neural classifiers	1. PP- Sharpening Filter, Hair removal using Dull Razor 2. FE- (i) Shape features extracted by ABCD rule (ii) Texture features by GLCM, Coarseness (iii) Color features extracted by Variance, Entropy, Skewness 3. FS&CLF-SVM and Neural classifiers	Risk level
[87]	Two way class Benign and Melanoma	ISBI 2016	SVM,RF,KNN, NB	1. SEG-Dice coefficient and achieved 77.5%(i) CNN based U-Net algorithm 2. FE- LBP, EH, HOG &Gabor. 3. CLF- SVM, RF, KNN, NB	85.19%
[74]	Benign and Melanoma	ISIC	SVM	1. PP-Median filtering (noise removal) 2. SEG&FE-DCT, HOG, Laws texture features, Entropy Filter 3. CLF-SVM	84.86%
[76]	Benign and Melanoma	University Medical Center Groningen Department of Dermatology	SVM, DT, KNN, RF	1. SEG -Otsu's thresholding, region growing, compression based methods, K-Means clustering. 2. FE-(i) Color Features by histogram; (ii) Texture using Gray Level Co-occurrence Matrix (iii) Border and asymmetry by solidity 3. CLF-SVM, DT, KNN, RF	89.07%

Table 4 (continued)

Ref	Diagnosis	Dataset	Classifier	Methodology	Accuracy
[62]	Melanoma and benign	MedNode	Ensembled Bagged Tree classifier	1. PP-Resize and De-noise by Gaussian Blur 2. SEG- Extraction of features from the lesion 3. FE-HSB + LBP 4.CLF- Ensembled Bagged Tree classifier	Accuracy = 95%, Sensitivity = 94%, Specificity = 97%, AUC = 99%
[58]	MEL, NEVI & SEK	ISIC2017	KNN	FE + KNN	Accuracy = 68%, Sensitivity = 80%, Specificity = 80%
[16]	7 class	ISIC2019	LDA	1. PP-Explore the dataset(80% train and 20% test) 2. CLF-S_{STF} statistical fractal signatures + LDA classifier	4-classes: Accuracy = 87%Sensitivity = 63%, Specificity = 89%, Precision = 65% 7-classes: Accuracy = 88%, Sensitivity = 41%, Specificity = 92%, Precision = 46%
[14]	Two class Melanoma, Non Melanoma	HAM10000	Linear SVM	1. PP-Increased brightness, addition of contrast 2. FE&FS- BHHO-S algorithm 2.CLF-linear SVM	Accuracy = 88%, Sensitivity = 89%, Specificity = 89% Precision = 86%

In [97], the author introduced a methodology for enhancing contrast in skin lesion images using a modified sigmoid transform based on the EfficientNetmodel. The segmentation results, measured by the Dice Similarity Index (DSI), ranged from 68% to 81% using Otsu's thresholding.

Lastly, [57] discussed a common model for skin cancer diagnosis utilizing CNNs, although further details about the specific approach and findings were not provided.

Overall, these studies demonstrate the effectiveness of CNN-based deep learning approaches in skin lesion analysis and showcase their potential for improving diagnostic accuracy in skin cancer detection and classification tasks.

The analysis of deep learning approaches is visualized in Table 5.

4.3 Hybrid strategies

Transfer learning, which makes use of CNN, built on large datasets like ImageNet, is the best option for skin cancer detection and classification [105]. By reusing the learned weights and features from these models, transfer learning enables improved performance and reduces training time. Fine-tuning the pre-trained models involves freezing and unfreezing specific layers to adjust the hyperparameters and keep ready for target.

In the field of dermoscopic imaging, both techniques ML and DL are widely employed. DL models, specifically deep neural networks (DNNs), are trained on annotated datasets containing benign and malignant melanoma images. Various datasets such as ISIC archive, HAM10000, PH2, Med Node, and DermIS are commonly used in skin cancer research [99].

CNN models have shown impressive results in classification, and segmentation in medical imaging applications [39]. Researchers have developed models using architectures like DenseNet121, VGG-16, AlexNet, ResNet-18, and EfficientNet, among others. These models are -trained on ImageNet and then fine-tuned for skin cancer tasks [25, 67].

Traditional ML approaches, such as SVM, have also been utilized for skin cancer diagnosis. SVM-based methods are used for texture analysis, color extraction, and classification [18].

Ensemble models, which combine multiple sources of information such as images, handcrafted features, and metadata for classification. These models use deep learning backbones like EfficientNet and employ techniques like feature extraction, followed by classification using multi-layer perceptrons (MLPs) or artificial neural networks (ANNs) [88].

In addition, studies have investigated DL models for threshold prediction and contrast enhancement in skin lesion analysis. Techniques such as deep CNNs and modified sigmoid transforms have been employed for accurate segmentation and improved visualization of skin lesions [15, 21, 98].

Overall, the research in skin lesion classification encompasses various approaches, including transfer learning, deep learning, traditional machine learning, ensemble models, and specialized techniques for specific tasks like threshold prediction and contrast enhancement. In (Table 6) the hybrid approaches for skin cancer detection has been visualized.

Experimental analysis using DL models was done to find the effective model for accurate classification. The analysis compared different DL models, such as VGG-16, ResNet-18, and EfficientNet, to identify with efficiency. The results confirmed the superior performance of DL models in accurately categorizing skin lesions based on various characteristics and features. This highlights the importance of utilizing DL techniques for reliable and effective skin cancer diagnosis.

Table 5 Deep learning approaches on skin cancer detection

Ref	Diagnosis	Dataset	Methodology	Accuracy (%)
[104]	7 Class	HAM10000	TL- InceptionV3, ResNet50 and Denset201 Transferring a pre-trained network's parameters to CNN	Accuracy = 96%
[37]	7 class	HAM10000	Custom CNN 1. PP-Augmentation of dataset 2. CLF-Dense-I and Dense-II (Custom CNN)	Accuracy = 96.3%, Recall = 96%, F1-score = 95.7%
[75]	2 class	ISIC archive	Custom shallow CNN Altering the brightness and contrast of images	Accuracy = 98.7%, Precision = 98.9%
[81]	7 class	HAM10000	DL approach (Pretrained)	Accuracy = 99.7%
[93]	2 class	ISIC2020	VGG16(New model based on VGG16) 90, 180, and 270 degree image rotation; centre cropping; brightness modification; and mirroring.	Accuracy = 87%, Sensitivity = 85.2%, F1 score = 92.2%, AUC = 92.3%
[26]	7 class	HAM10000	Custom CNN-AlexNet	Accuracy = 87.8%, Specificity= = 96.2%, Precision = 78.7%, Recall = 77.4%, F1 score = 77.8%
[89]	2 class	HAM10000	VGG16, VGG19, DenseNet101& ResNet101	Accuracy = 84.8%
[80]	2 class	HAM10000, ISIC2019, ISIC2020	Custom CNN'son Pretrained network (VGG16, ResNet50, DenseNet169 and EfficientNetB3)	Accuracy =98%, Specificity = 99.9%, Precision = 96.1%, Recall = 98%, F1 score = 97%, AUC = 70.9%
[5]	2 class	ISIC archive	Zooming, vertical/horizontal flipping, shearing, and rescaling TL use on MobileNetV2, DenseNet201, XceptionNet, and ResNet50	Accuracy =86.6%, Precision = 86.5%, Recall = 86%, F1score = 86.2%
[68]	8 class	ISIC2019	DNN with specific	Accuracy =95.6%, Specificity = 96.3%, Precision =84.7%, Recall = 92.5%, F1score = 88.4%
[56]	2 class	2742 dermoscopic images from ISIC dataset	Region-based CNN with ResNet152 Region-based CNN and masks was employed for the region of interest extraction, and ResNet152 applied for classification.	90.4% 82%- Sensitivity 92.5%-Specificity

Table 5 (continued)

Ref	Diagnosis	Dataset	Methodology	Accuracy (%)
[30]	Multi-class classification	PH^2, HAM10000	Resnet50, VGG16 F_{EX}-Resnet50, VGG16 and Deeplabv3 + modified LSTM	PH^2: Accuracy =94%, Sensitivity = 94%, Specificity = 93%, Precision = 90%, F1 score = 92%; HAM: Accuracy = 94%, Sensitivity = 94%, Specificity = 94%, Precision = 34%, F1score = 50%
[61]	Multi-class classification	HAM10000	Custom CNN	Accuracy = 87%, Sensitivity = 86%, Precision = 87%, F1score = 86%
[85]	Multi-class	ISIC2019, HAM10000	Fine-tuned models (Ensemble) (ResNext, SeResNext, ResNet, Xception and DenseNet)	Accuracy = 87%, Precision = 87%, Recall = 93%, F1score = 89%, MCC = 87%
[45]	Multi-class:	ISIC2017, ISIC2018, ISIC2019	Custom CNN Horizontal/vertical flipping,12.5% and rotation (30 to 30 degrees)	ISIC2017:Accuracy = 93%, Sensitivity = 93%, Specificity = 91%, Precision = 94%, F1score = 93%, AUC = 96% ISIC2018:Accuracy = 89%, Sensitivity = 89%, Specificity = 96%, Precision = 90%, F1score = 89%, AUC = 99%; ISIC2019:Accuracy = 90%, Sensitivity = 90%, Specificity = 98%, Precision = 91%, F1score = 90%, AUC = 99%
[44]	2 class	ISIC archive, ISIC2016, MedNode, PH2	ModifiedPre-trained MobileNetV2	ISIC archive: Accuracy = 85%, Sensitivity = 85%, Specificity = 85%, Precision = 83% ISIC2016:Accuracy = 83%, Sensitivity = 36%, Specificity = 95%, Precision = 64%MedNode:Accuracy = 75%, Sensitivity = 76%, Specificity = 73%, Precision = 67%PH2: Accuracy = 72%, Sensitivity = 33%, Specificity = 92%, Precision = 67%

Table 5 (continued)

Ref	Diagnosis	Dataset	Methodology	Accuracy (%)
[40]	2 class	PH2 MedNode DermIS and Quest ISIC archive (2016,2017 and 2018)	Refined Residual Deep Convolutional Neural Network	PH2: Accuracy = 94.97%, Sensitivity = 92.32%, Specificity = 96.21% MedNode: Accuracy = 92%, Sensitivity = 91.99% Specificity = 91.99%
[41]	7 class	ISIC2018	Transfer learning with AlexNet	Accuracy = 98.7%, Sensitivity = 95.6%, Specificity = 99.2% Precision = 95%
[42]	3 class	MED NODE DermIS and Quest ISIC2017	DCNN (AlexNet, ResNet101, GoogleNet)	Modified GoogleNet MED NODE Accuracy = 99.29% DermIS and Quest Accuracy = 99.15% ISIC2017 Accuracy = 98.14%

2 class- Benign and Malignant

3 class - Melanoma, Seborrheic Keratosis, and Nevus

7 class-NEVI/AN (Atypical NEVI), COM NEVI (Common Nevi), MEL (Melanoma), SEK (Seborrheic keratosis), BACC (Basal cell carcinoma), DEF (Dermato fibroma), ACK (Actinic keratosis), VASCL (Vascular lesion)

8 class-SQCC (Squamous cell carcinoma), NEVI/AN (Atypical NEVI), COM NEVI (Common Nevi), MEL (Melanoma), SEK (Seborrheic keratosis), BACC (Basal cell carcinoma), DEF (Dermato fibroma), ACK (Actinic keratosis), VASCL (Vascular lesion)

Table 6 Hybrid approaches on skin cancer detection

Ref	Diagnosis	Dataset	Methodology	Accuracy
[88]	7 Class	ISIC2018,ISIC2019	FE- hand-extracted features + ANN and B4 to 486 B7 EfficientNet networks	ISIC2018: Accuracy = 91%, Sensitivity = 98%, ISIC2019:Accuracy = 86%, Sensitivity = 98%
[18]	2 class	HAM10000	FE- pre-trained DenseNet121 + XGBoost	Accuracy = 91%; Sensitivity = 93%; Precision = 91%; F1- score = 91%
[15]	2 class	HAM10000, PH^2	FE-from ResNet and 503 and hand crafted, EfficientNet + ANN	HAM10000: Accuracy = 95%, Sensitivity = 95%, Specificity = 95%, Precision = 95%, F1score = 95%; PH2: Accuracy = 98%, Sensitivity = 98%, Specificity = 98%, Precision = 96%, F1 score = 97%
[34]	2 class	ISIC2016, ISICI2017, HAM10000, PH^2	FE-Pre-trained DenseNet121 network + MPL	ISIC2016:Accuracy = 81%ISICI2017:Accuracy = 81%; HAM10000: Accuracy = 81%; PH2: Accuracy = 98%

Table 7 Hyper-tuning parameters-ML classifiers

ML classifier	Hyper tuning parameter
Linear discriminant analysis	solver = 'svd'
Logistic regression	random_state = 9
K-nearest neighbor	n_neighbors = 5
Decision tree	estimators = 100
Random forest	n_estimators = 200 and random_state = 0
Support vector machine	kernel = linear,c = 1,and random_state = 0

4.4 Experimental analysis and discussion

4.4.1 ML algorithms with HAM10000 dataset

The proposed study employed various ML models, including DT, RF, SVM, KNN, LOR, and LDA. These models were evaluated using different evaluation metrics to compare their performance. The hyper-parameters used by these ML algorithms are provided in Table 7 [90].

The accuracy of the model was analyzed using two approaches: with data augmentation and without data augmentation. The findings are in Table 8.

Random Forest approach outperforms other ML algorithms. Table 9 displays experimental outcomes for machine learning, including accuracy, precision, recall, and F1-score [90].

4.4.2 Fine-tuned pre-trained model with HAM10000 dataset for skin lesion classification

VGG-19 The VGG-19 is a CNN architecture with 19 layers. The network is capable of classifying 1000 different categories. The input images used for training are 224×224 pixels in size. The VGG-19 architecture is known for its simplicity, achieved by stacking multiple 3×3 convolutional layers and reducing volume size using maximum pooling. A softmax layer and two fully linked layers with 4096 nodes each follow the network. Convolutional and completely linked layers represented by the "19" in VGG-19.

In the context of skin lesion classification, the VGG-19 model was applied to the HAM10000 database. The dermoscopic images undergo preprocessing steps such as resizing, frame removal, and cropping of the skin lesion. The preprocessed images are then input into the VGG-19 model for classification. The model is trained and validated using evaluation metrics derived from the training dataset.

Table 8 Accuracy –ML classifiers

Dataset	Accuracy (%)					
HAM10000	LDA	LR	KNN	DT	RF	SVM
Without augmentation	35.18	48.92	41.78	44.28	58.57	42.14
With augmentation	57.58	58.12	48.39	68.66	87.32	53.12

Table 9 Evaluation Metrics-ML model

Evaluation (%)	ML Model					
	RF	DT	LOR	LDA	SVM	KNN
Accuracy	87	68	58	57	53	48
Precision	94	75	56	62	54	54
Recall	94	74	55	58	50	50
F1 Score	94	74	55	54	50	50

VGG-16 CNN architecture that extensively employs convolution layers continued by pooling layers throughout its structure. It culminates in 2 FC layers and a softmax, creating a seamless connection. The number "16" in VGG-16 represents the total weighted layers present in the architecture. With a staggering 138 million parameters, the VGG-16 network is considered large and complex, capable of capturing intricate patterns and features in the data [91].

MobileNet model The MobileNet network is composed of 28 layers. By adjusting the width multiplier hyper-parameter, the number of parameters in the standard MobileNet architecture, which is around 4.2 million, can be further reduced. In classification, the skin lesion image undergoes pre-processing to remove duplicates, and then it is processed with MobileNet for classification [19, 84].

Xception The Xception architecture, also known as extreme inception, incorporates filtering techniques that consider the cross-channel or depth dimension as well as the spatial dimension. This involves scanning a 2×2 pixel patch across all RGB channels using convolution filters similar to an image input layer. The initial input is projected onto smaller input regions using 1×1 convolutions, and different filters are applied to modify these smaller 3D data blocks. Spatial correlation maps are generated for each output separately, followed by an 11-depth convolution to capture cross-channel correlation. This approach separates the learning of cross-channel and spatial information by depth, reducing computational costs and memory requirements. The Xception architecture consists of sixteen hidden levels distributed among fourteen modules [63].

An ensemble model is used for classification of skin lesion by combining multiple models for improved categorization. Figure 7 illustrates the ensemble model consisting of Xception, ResNet152, and InceptionV3 architectures for skin cancer detection [4].

The detailed model described above, which includes architectures such as VGG-19, MobileNet, and Xception, is combined with the Skin Cancer dataset for the purpose of identifying a suitable pre-trained model for detection. By training and evaluating these pre-trained models on the skin cancer dataset, their performance in accurately detecting and classifying skin cancer can be assessed. This process helps in determining which model performs best for skin cancer detection based on the specific characteristics and requirements of the dataset.

4.4.3 Performance metrics

In deep learning, common metrics like accuracy and loss are to evaluate the performance of a model.

The model's accuracy is gauged by how well it can diagnose the classes of data samples. Based on the values of TP, TN, FP, and FN, it is calculated.

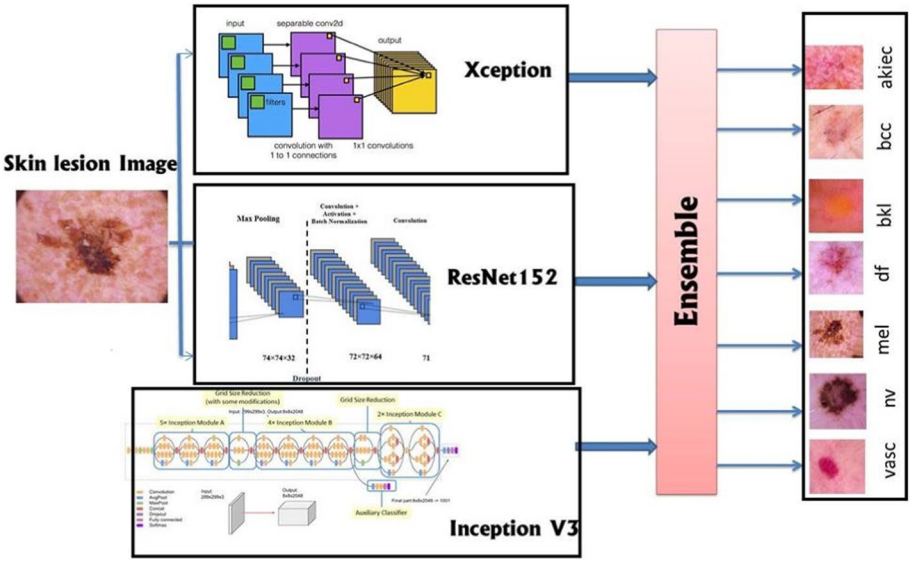


Fig. 7 Ensemble model (Xception, ResNet152, and inceptionv3)

An indicator, the model matches the training data is the training loss. It is determined by adding the mistakes for each training set example. The training loss helps assess how the model is learning and adjusting its weights and biases during the training process.

Similar to this, the validation loss is a statistic used to assess how well the model performs on a different validation set. Additionally, the errors for each sample in the validation set are added up to determine the validation loss.

Implications of loss Both the training loss and the validation loss give information about how well the model generalises to new data and can show whether the training data are being overfitted or underfitted by the model. Monitoring these data allows for performance improvements in the model architecture, hyperparameters, and training procedure.

Confusion matrix Summary of a classification model's performance by identifying the count of instances of true positives (TP), false negatives (FN), false positives (FP), and true negatives (TN). From the confusion matrix, important metrics can be derived.

The true positive rate (TPR), also known as recall and sensitivity measures, indicates the percentage of the positive class that is correctly classified, and the formula has been shown in Eq. (1) and followed by false negative rate (FNR),Specificity, false positive rate (FPR) in Eqs. (2, 3, 4) respectively.

$$Sensitivity/TPR/Recall = TP/(TP + FN) \tag{1}$$

False negative rate, the proportion of the positive class got classified incorrectly.

$$FNR = FN/(TP + FN) \tag{2}$$

The percentage of negative classes that are accurately identified allows one to determine the specificity of a model.

$$Specificity = TN/(TN + FP) \tag{3}$$

False positive rate, the proportion of negative classes got classified incorrectly.

$$FPR = FP/(TN + FP) = 1 - Specificity \tag{4}$$

The performance metrics for the classification also includes ROC and AUC curves. ROC based on the false positive rate or specificity and true positive rate or sensitivity at various thresholds of classification scores. General format for confusion matrix are shown in (Fig. 8).

4.4.4 Performance of fine-tuned pre-trained model with HAM10000 dataset

Hyperparameters play a crucial role in fine-tuning pre-trained models. These parameters are adjustable settings that influence the learning process and overall performance of the model [101]. In Table 10, relevant hyperparameters for the procedure are listed, specifying the values chosen for each model.

Fine-tuning pre-trained models involves adjusting hyperparameters that are carefully selected and tuned to achieve optimal performance for the specific task. The identified values for these hyperparameters can significantly impact the model's training process and final results.

Optimizer Optimizers are an essential component in deep learning for modifying a model's parameters during training. Their main purpose is to optimize the model's performance by adjusting the weights to minimize the loss function [10]. In deep learning, various optimizers can be used, each with its own algorithm and update rules. Some commonly used optimizers include:

1. Adam: Adam has the advantages of RMSprop and momentum methods. It provides effective optimisation by adjusting the learning rate based on gradients [96].

Fig. 8 Confusion matrix

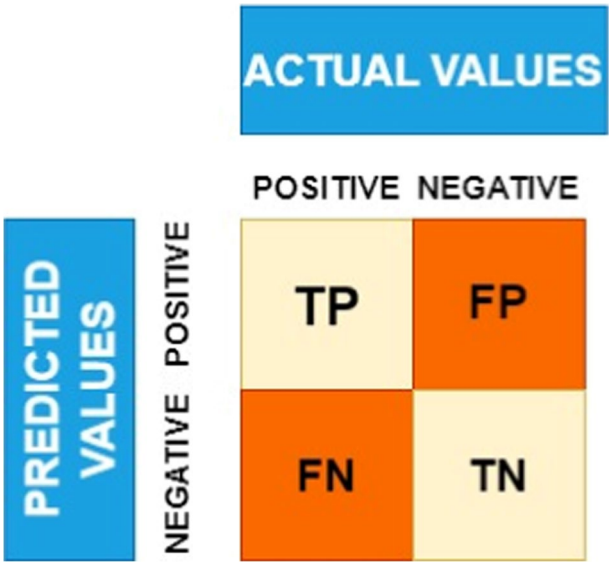


Table 10 Hyper tuning parameters for fine-tuned pretrained model [102]

Fine _tuned Pretrained Model	Size of the input image	Learning rate	Optimizer	Epoch	Batch size
VGG19	(224,224,3)	0.0001	Adam	35	64
VGG16	(224,224,3)	0.0001	Adam	35	64
DenseNet	(224,224,3)	0.0001	Adam	30	32
ResNet152	(224,224,3)	0.0001	Adam	35	64
InceptionV3	(224,224,3)	0.0001	Adam	35	64
InceptionResNetV2	(224,224,3)	0.0001	Adam	25	64
Xception	(224,224,3)	0.0001	Adam	35	64
MobileNetV2	(224,224,3)	0.01	Adam	15	10

2. Momentum: Momentum optimizer uses a moving average of past gradients to accelerate the learning process. It accumulates a velocity term that directs the update direction, helping the optimizer to overcome local optima [96].
3. RMSprop (Root Mean Square Propagation): The learning rate is divided by a moving average of the root mean square of previous gradients in RMSprop. It helps to stabilize and normalize the learning process, especially in cases where the gradients vary significantly [96].

These optimizers offer different strategies for updating the model's weights during the training process, allowing for efficient convergence and better generalization. The choice of optimizer depends on the specific problem, data, and network architecture and it can impact the training speed and final performance of the model.

Adam (adaptive moment learning rate) optimizer Adam combines the advantages of two other optimization methods: momentum and RMSprop. Based on estimations for the first and second moments of the gradients, it adjusts the learning rate for each parameter. The optimizer converges more quickly and successfully manages various types of data and model architectures because to its adjustable learning rate.

The update equation for Adam can be expressed as follows:

$$\boldsymbol{\theta(t+1) = \theta(t) - \alpha * m(t) / \left(sqrt(v(t)) + \varepsilon \right.}$$

where:

$\theta(t)$ => parameter at iteration t,

α => learning rate,

$m(t)$=> estimate of the mean,

$v(t)$=>estimate of the variance,

ε=>a tiny constant to prevent division by zero.

Changing the learning rate in accordance with the gradient's first and second moments, Adam optimizes the weight updates, making it a widely used optimizer in deep learning.

The model's total parameters (TtP), trainable (TrP) and non-trainable parameters (NTrP) are shown in the Table 11.

The HAM10000 dataset was used to train the fine-tuned pre-trained model, and the accuracy and loss were calculated and displayed in the Table 12.

Table 11 Trainable parameters for fine-tuned pre-trained model

Fine _tuned pretrained model	Total parameters (TtP)	Trainable parameters (TrP)	Non trainable parameters (NTrP)
VGG19	20,290,631	20,290,631	0
VGG16	14,980,935	7,345,671	7,635,264
DenseNet	19,309,127	8,097,479	11,211,648
ResNet152	59,423,623	59,272,199	1,51,424
InceptionV3	22,855,463	22,821,031	34,432
InceptionResNetV2	55,127,271	24,352,647	30,774,674
Xception	21,914,159	21,859,631	54,528
MobileNetV2	3,236,039	3,214,151	21,888

Based on the findings presented, it can be concluded that MobileNetV2 and the ensemble of DenseNet and InceptionV3 models have shown superior performance compared to other models in terms of effectiveness. These models have demonstrated higher accuracy or better classification results in the given context of skin lesion classification or detection. The use of MobileNetV2 and the ensemble of DenseNet and InceptionV3 can be recommended as effective choices for skin lesion analysis tasks.

4.4.5 CNN (convolutional neural network) MODEL with HAM10000 dataset

The CNN architecture consists of several key components:

1. Convolution Layer: This layer uses filters or kernels to extract features from the input image through a process called convolution. Each filter detects specific patterns or features in the image, resulting in feature maps [78].

Table 12 Performance on fine-tuned pre-trained model

Fine _tuned pre-trained model	Training		Validation		Testing	
	Accuracy (%)	Loss	Accuracy (%)	Loss	Accuracy (%)	Loss
VGG19	86.93	0.3617	84.38	0.492	82.87	0.520
VGG16	88.92	0.3140	79.46	0.602	79.82	0.650
DenseNet	98.00	0.0580	85.86	0.579	83.93	0.691
ResNet152	95.83	0.1080	80.21	0.870	81.77	0.880
InceptionV3	98.04	0.0599	85.42	0.655	83.97	0.690
InceptionResNetV2	93.32	0.1898	81.75	0.617	82.53	0.640
Xception	98.60	0.0392	86.72	0.676	84.57	0.800
MobileNetV2	94.22	0.1936	88.81	0.331	87.71	0.350
Ensemble (DenseNet, InceptionV3)	–		88.80	0.399	88.52	0.411
Ensemble (Xception,ResNet152, InceptionV3)	–		87.53	0.443	85.92	0.487

2. Kernel or Filter: The kernel is an iterative matrix that performs dot product operations with subregions of the input data. It helps extract image features and is typically of size 3×3 or 5×5. Different kernels are applied to capture local features in the image [78].
3. Padding: Padding is used to preserve data at the borders of the activation maps during convolution. It ensures that the spatial dimensions are maintained and helps in building deeper networks. Zero-padding is commonly used for its simplicity and effectiveness [78].
4. Stride: The stride determines how many pixels the kernel moves with each operation. It affects the output layer dimensions and feature extraction. A stride of 1 means the kernel moves one pixel at a time. Increasing the stride value reduces the output dimensions and restricts feature extraction [78].
5. ReLU Layer: The Rectified Linear Unit (ReLU) layer applies an element-wise activation function to introduce non-linearity. It replaces negative activations with zero, enhancing the features of the image [60].
6. Fully Connected Layer: This layer takes the output of the convolutional process and performs classification based on the extracted features. It calculates class scores for each image and assigns the predicted class based on the highest probability score [60].

In addition to these layers, CNNs may also include other components such as dropout layers and activation functions to further enhance performance. The CNN architecture processes the pre-processed image and classifies it into different categories. The classification process involves both pre-processing and classification steps. The overall workings of the CNN model for skin lesion classification are depicted in Fig. 9 [60].

The experimental analysis involved adjusting the number of convolution layers and fine-tuning hyperparameters to evaluate the CNN's performance. The model's hyperparameters were tuned to optimize its performance. The model was trained for a total of 25 epochs, as indicated in Table 13. Initially, the model was trained with 50 epochs and a learning rate of 0.001. After 16 epochs, the validation accuracy started to decrease. At epoch 20, the training process was stopped, and the learning rate was reduced as it reached the patience value. The model then continued training with the reduced learning rate and achieved its best validation accuracy at epoch 22. However, the model's

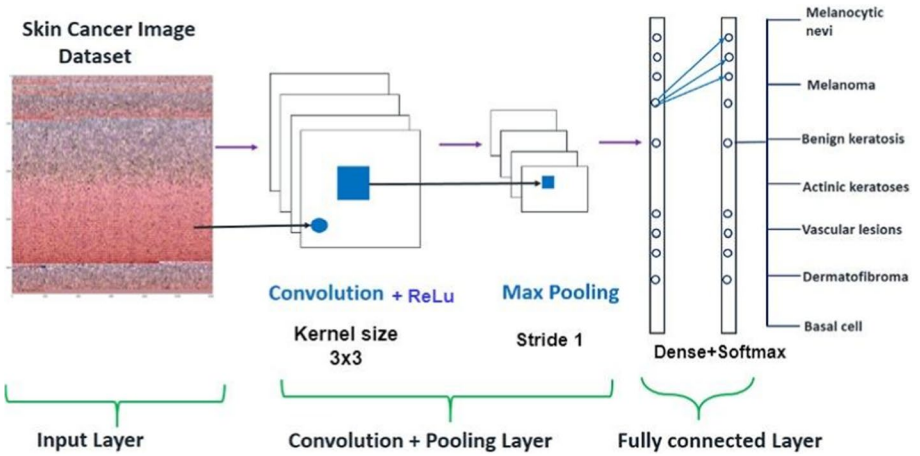


Fig. 9 Skin lesion classification using CNN Model [78]

Table 13 Hyper tuning and trainable parameters of CNN models

CNN model	Total parameters	Trainable parameters	Non -trainable parameters	Hyper tuning parameters			
				Epochs	Batch size	Lr-rate	Optimizer
CNN(7 layered)	1,275,079	1,273,671	1408	25	128	0.001	Adam
CNN(5 layered) with Batch Normalization	6,19,207	6,17,799	1408	25	128	0.001	Adam
CNN(5 layered) without Batch Normalization	4,11,400	4,11,400	0	25	64	0.001	Adam
CNN(3 layered)	2,124,839	2,124,839	0	30	64	0.001	Adam

accuracy gradually declined, and early stopping was enabled when no further improvement was observed. The model weights were restored to the best epoch, and the training process concluded at epoch 25. Table 13 provides details of the hyperparameters used for hyper-tuning the CNN model.

After training the model with the chosen hyperparameters, the accuracy and loss values were determined and visualized in Table 14. This table presents the accuracy and loss metrics obtained during the training process, providing insights into the model's performance on the dataset.

The results indicate that when using the HAM10000 raw dataset, the CNN model's validation accuracy does not exceed 80%. Additionally, the models with 7 and 5 layers exhibit overfitting, as evidenced by the significant difference between the training loss and validation loss. However, the CNN model with 3 layers performs better in terms of avoiding overfitting. It achieves a more balanced performance between training and validation, resulting in improved accuracy on the raw dataset.

In the context of skin cancer classification, dealing with imbalanced datasets is crucial for accurate model training. Skin cancer datasets often exhibit class imbalance, where certain types of skin lesions are more prevalent than others. This class imbalance can lead to biased models with lower accuracy on minority classes.

To solve this problem, particular methods can be used to balance the dataset and enhance the model's performance. One approach is data augmentation, which involves generating new synthetic samples by applying various transformations to the existing data.

Impact of data augmentation Techniques for image augmentation, such as rotation, scaling, flipping, and brightness or contrast adjustments, can be used to classify skin cancer. These modifications add more diverse samples and aid in balancing the distribution of classes. One effective method is random rotations, which introduce variations in the orientation of the images and help the model generalize better to different angles and viewpoints [12].

Two common approaches are used to fit the input size of the augmented data:

Table 14 Performance on CNN models

CNN model	Size of input image	Training		Validation		Testing	
		Accuracy (%)	Loss	Accuracy (%)	Loss	Accuracy (%)	Loss
CNN(7 layered) with Batch Normalization	(28,28,3)	99.95	0.0082	77.92	0.7192	76.94	0.8204
CNN(5 layered) with Batch Normalization	(28,28,3)	98.25	0.0807	77.22	0.8191	76.55	0.8508
CNN(5 layered) without Batch Normalization	(28,28,3)	91.22	0.2572	73.80	0.8804	75.88	0.870
CNN(3 layered)	(64,64,3)	79.85	0.5263	77.48	0.6034	76.54	0.6466

(i) Scaling: The resolution of the images is multiplied by a scaling factor to resize them. This method allows for adjustments in the overall size of the images while preserving their spatial content [12].
(ii) Cropping: This approach involves extracting sub-regions from the images. The cropping can be done by selecting either the center portion or a random area of the image. This technique helps maintain the important features of the image while reducing its size [77].

By applying data augmentation techniques, the training process benefits from a larger and more diverse dataset. This helps address the issue of overfitting. The augmented dataset provides the model with a wider range of examples, enabling it to learn more robust and generalizable representations of the skin lesion classes.

Performance of fine-tuned pre-trained and CNN model after augmentation Tables 15 and 16 shows the impact of augmentation in the model, the accuracy of the model was improved when compared to the raw dataset.

The table illustrates the improvement in model performance in terms of accuracy when using data augmentation techniques on an imbalanced dataset. These results highlight the significance of augmenting the dataset to address class imbalances. The experimental analysis demonstrates that CNN models with different layer sequences perform well when trained on a balanced dataset.

Among the models evaluated, the fine-tuned pre-trained model MobileNetV2 shows efficient performance compared to other models. Additionally, when using a custom CNN model, CNN architectures with 3 or 5 layers yield favorable results compared to other CNN configurations. These findings are consistent with previous studies conducted using similar models, validating the effectiveness of the proposed approach.

Comparing results with literature Table 17 and Fig. 10 show the comparative results with previous experiments.

Advantages of DL Techniques Deep learning models, including pre-trained models, transfer learning models, and CNN-based models, offer several advantages in image recognition tasks. They can automatically extract pertinent characteristics from images. Transfer learning allows leveraging pre-trained models for specific tasks, reducing the need for extensive training. Deep learning models achieve high accuracy and can process large amounts of data efficiently, making them suitable for image recognition. They also enable end-to-end learning, simplifying the workflow. Overall, deep learning has revolutionized image recognition with its efficient and effective techniques.

5 Challenges

5.1 Extensive background

Challenges in Employing Neural Networks for Skin Cancer Diagnosis

1. Analysis and Comprehension of Dermoscopic Images: One of the major difficulties in using neural networks for skin cancer diagnosis is the task of analyzing and comprehending the characteristics of dermoscopic images. This process requires significant time and effort.

Table 15 Performance of fine-tuned pre-trained model after augmentation

Fine _tuned pretrained model with augmentation	Training		Validation		Testing	
	Accuracy (%)	Loss	Accuracy (%)	Loss	Accuracy (%)	Loss
VGG19	87.90	0.3285	87.76	0.4084	87.76	0.4083
VGG16	87.31	0.3448	86.51	0.4548	86.50	0.4548
DenseNet	99.87	0.0450	96.82	0.0220	96.80	0.0225
ResNet152	99.90	0.0120	96.71	0.0240	96.71	0.0241
InceptionV3	99.90	0.0110	97.01	0.0230	97.00	0.0229
InceptionResNetV2	97.32	0.0298	91.75	0.1170	91.55	0.1260
Xception	99.60	0.0192	96.72	0.0667	95.27	0.0567
MobileNetV2	97.99	0.0671	97.58	0.0551	97.28	0.0591
Ensemble (DenseNet, InceptionV3)	–		97.50	0.0561	97.02	0.0610
Ensemble (Xception,ResNet152, InceptionV3)	–		96.53	0.0340	95.92	0.0420

Table 16 Performance of CNN Model after augmentation

CNN model with augmentation	Size of the input image	Training		Validation		Testing	
		Accuracy (%)	Loss	Accuracy (%)	Loss	Accuracy (%)	Loss
CNN(7 layered) with Batch Normalization	(28,28,3)	99.98	0.0047	91.38	0.4199	92.08	0.3631
CNN(5 layered) with Batch Normalization	(28,28,3)	99.56	0.0385	89.62	0.3734	89.38	0.3930
CNN(5 layered) without Batch Normalization	(28,28,3)	100	0.0002	97.72	0.1313	98.04	0.1100
CNN(3 layered)	(28,28,3)	96.83	0.0895	98.02	0.240	97.35	0.280

Table 17 Comparative of results with previous experiments

HAM 10000 Dataset	Model	Accuracy
Balanced	Fine- tuned MobileNetV2	97.58
Not Balanced		88.81
Not Balanced	Indraswari, R.et al. [44]	85.00
Balanced	Chaturvedi, S. et. al [19]	95.84
Balanced	Customized CNN	98.02
Not Balanced		74.17
Balanced	Shetty, B.et. al [90]	95.18
Balanced	Nugroho, A. et. al [79]	80.00

2. Variation in Lesion Size: The variation in lesion size poses another challenge in skin cancer identification. The accuracy of diagnosis can vary depending on the size of the lesion. Smaller lesions, particularly those measuring 1 mm or 2 mm, make diagnosis more challenging and prone to errors.
3. Unbalanced Datasets: Unbalanced datasets pose challenges in training neural networks for skin cancer diagnosis. These datasets often have a highly uneven distribution of photos for different types of skin cancers. While there may be numerous images of common carcinomas, there are only a few images of rare forms, making it difficult to generalize visual characteristics from dermoscopic images.
4. Analysis of Genetic and Environmental Factors: When combined with environmental hazards like excessive UV exposure, these genetic factors significantly increase the likelihood of cancer. Incorporating these factors into deep learning algorithms can improve their performance [92].

6 Discussion, conclusion and future enhancement

To identify the course of therapy and increase the likelihood that a patient will survive, early detection and proper diagnosis of skin lesions are essential, especially in situations of malignant lesions. However, manual diagnosis by dermatologists can be time-consuming and challenging. To simplify and expedite this process, Computer-Aided Diagnosis (CAD) technologies have been employed.

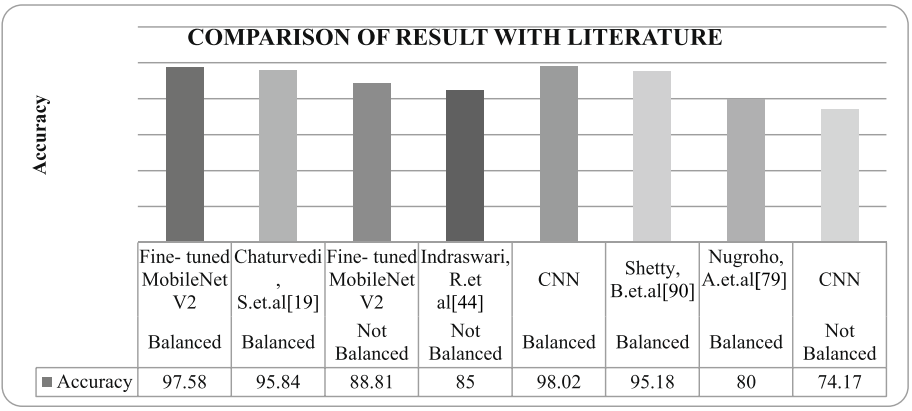


Fig. 10 MobileNetV2 and CNN with previous study

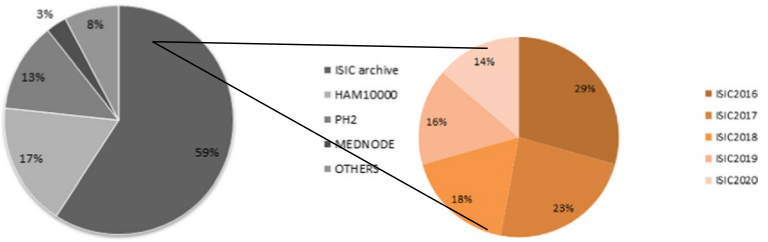


Fig. 11 Dataset analysis. Others–DermQuest, DermIs, 7 point checklist, DermNZ, and IDS datasets

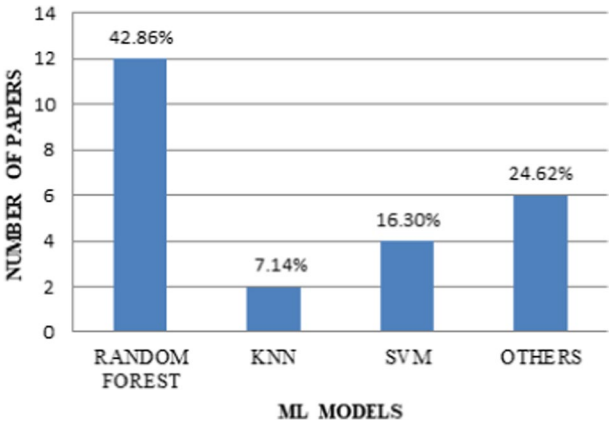
This review aims to provide an overview of the application of deep learning (DL) and machine learning (ML) in dermatology. Academic works focusing on ML and DL methods for identifying and classifying skin lesions were selected from publications between 2012 and 2022. The search was conducted in databases such as arXiv and Science Direct, resulting in the inclusion of approximately 70 articles. These articles utilized photos from open datasets and reported the model performance in terms of classification accuracy.

It is worth noting that no publications were included before 2016, as the inclusion criteria required the use of public datasets. Furthermore, over 50% of the articles on DL and ML/DL were published from 2021 onwards, indicating a recent surge in research activity in this field. Additionally, within the selected articles, around 70% were published within the last two years, highlighting the increasing interest and advancements in DL and ML for skin lesion classification.

6.1 Dataset analysis

According to the study mentioned in this review article, the datasets most frequently used for training and testing classification models in the cited studies were HAM10000 and ISIC archive. These datasets have been widely utilized in various research papers to develop and evaluate skin lesion classification models. The availability and quality of these datasets

Fig. 12 ML model analysis



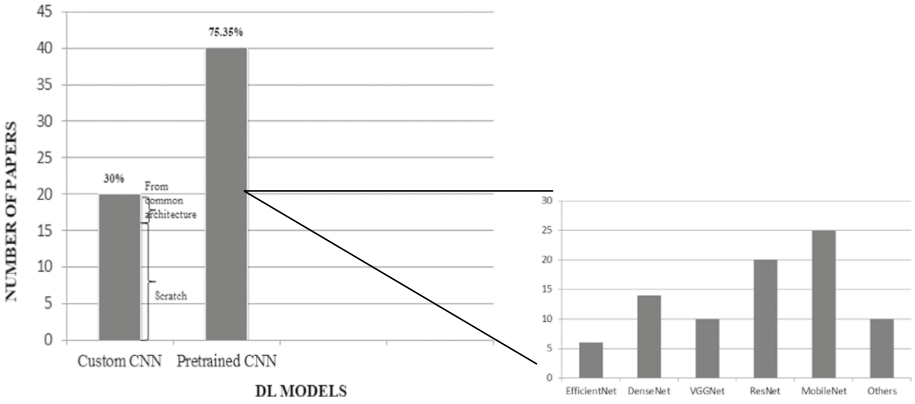


Fig. 13 DL model analysis

make them popular choices for training deep learning and machine learning algorithms in the field of dermatology. The analysis is visualized in Fig. 11.

6.2 Algorithm analysis

According to the study presented in the review article, pre-trained convolutional neural networks (CNNs) were found to be the dominant deep learning (DL) and machine learning with DL (ML/DL) technique for skin lesion classification (Fig. 12). These pre-trained CNN models have shown excellent performance in extracting meaningful features from skin lesion images and achieving high accuracy in classification tasks.

On the other hand, random forest (RF) models emerged as the most popular machine learning (ML) classifiers in the studies (Fig. 13). RF models have been widely utilized in skin lesion classification tasks, showcasing their effectiveness in handling structured data and providing good predictive performance.

However, among the various solutions examined, DL-based approaches, particularly deep CNNs, were found to be prevalent in addressing the categorization and diagnosis of skin lesions. The ability of deep CNNs to automatically learn intricate patterns and representations from images contributes to their potential for increasing the efficiency and reliability of skin lesion classification and diagnosis tasks.

The review conducted in this study explored different ML and DL algorithms for the diagnosis and classification of skin lesions. The study focused on ML techniques and deep learning strategies that combined CNN with other pre-trained models to categorize skin lesion images.

Every algorithm has benefits and limitations, and selecting the appropriate categorization technique is crucial for achieving optimal results. However, the review findings suggest that CNN-based approaches, particularly when integrated with pre-trained models, tend to outperform ML methods in terms of accuracy and effectiveness. DL algorithms, including CNN, show promise in delivering more robust and reliable results.

In the future, further research and exploration of different DL algorithms will be conducted to enhance the efficiency and performance of the models. Specific focus can be given to key stages such as segmentation and classification, where advancements in deep learning techniques can lead to improved outcomes in skin lesion diagnosis and classification.

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